Reda Rafi

Senior Software Engineer - Python - Distributed Systems - Cloud Solutions
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PROFESSIONAL SUMMARY

Versatile Software Developer with 8+ years of experience in software development, backend engineering, **microservices architecture**, and distributed systems. Strong proficiency in **Python**, **FastAPI**, **Flask**, **SQL**, **MySQL**, and **AWS**, with a proven track record of designing secure, scalable distributed systems and optimizing performance to meet SLAs. Skilled in end-to-end delivery of complex 3+ month initiatives, conducting peer code reviews, and mentoring junior engineers. Experienced in **Docker** containerization, **RabbitMQ**, and collaborative cross-functional technical design.

CORE TECHNICAL SKILLS

Programming: Python, SQL, PL SQL, Flask, FastAPI, Object-Oriented Programming (OOP)
Architecture & Systems: Microservices Architecture, Distributed Systems, RESTful APIs, Message Brokers (RabbitMQ, Kafka, Kinesis)
Big Data & Processing: PySpark, Apache Spark, Spark SQL, Hadoop, Large-Scale Data Processing
ETL & Data Processing: ETL/ELT, Data Pipelines, Data Transformation, Data Modeling, Data Quality
Databases & Data Stores: MongoDB, Amazon Redshift, Amazon Athena, Amazon S3, BigQuery
Processing Optimization: Partitioning, Data Pruning, Efficient Joins, Performance Optimization, Cost/Performance Optimization
Databases & RDBMS: MySQL, MongoDB, Amazon Redshift, Amazon Athena, Transactions, Isolation Levels, Locking, Deadlocks
Cloud & DevOps: Docker, AWS (Glue, Lambda, S3, Redshift, Athena, EventBridge, SQS, CloudWatch), CI/CD, Linux/Unix Environments
Collaboration & Practices: GitHub, Git, Code Reviews, Mentorship, Agile/Sprint Planning, Technical Design Specifications
Python Libraries: Pandas, NumPy
Streaming: Kafka, Kinesis, Event-Driven Pipelines
DevOps & Infrastructure: Docker, CI/CD, CloudFormation
BI & Analytics: Looker, Tableau, Power BI, Google Data Studio, Kibana

PROFESSIONAL EXPERIENCE

Senior Python Developer

SOFRECOM – Rabat, Morocco | Sep 2024 – Present

- Develop and maintain **Python-based solutions** supporting scalable analytics and reporting platforms.
- Lead the migration of historical data from **ELK to MongoDB**, designing scalable data structures to establish a reliable Source of Truth.
- Design and implement standardized data models and processing workflows to improve traceability and governance.
- Develop backend services using **Python, Flask, and FastAPI** to expose datasets and support platform integrations.
- Build and optimize workflows for processing and transforming large volumes of data.
- Implement data validation and quality practices to improve reliability of production datasets.
- Develop reporting datasets, dashboards, and analytical solutions supporting customer engagement and campaign performance.
- Translate business requirements into scalable **Python-based workflows and production-ready solutions**.
- Automate reporting and data processes to improve operational efficiency and reliability.

- Design and deliver high-quality features, own end-to-end delivery of large features (3+ month initiatives), create technical design specifications collaboratively, participate in sprint planning with accurate estimates, optimize system performance to meet SLAs, conduct peer code reviews, and mentor junior engineers.

Python / AWS Developer

CAPGEMINI – Casablanca, Morocco | Sep 2021 – Sep 2024

- Designed and developed scalable **Python-based ETL and data processing workflows** on AWS.
- Developed processing solutions using **Python, SQL, PySpark, and Apache Spark** for large-scale data transformation.
- Developed scalable Python-based **microservices** architecture and distributed systems.
- Worked with **Docker** and container-based systems in **Unix/Linux environments** using **GitHub** for code repositories.
- Built and optimized datasets for reporting, analytics, and dashboard visualization.
- Implemented data quality checks, validation, monitoring, and logging to improve application and workflow reliability.
- Applied optimization techniques including **partitioning, pruning, and efficient joins** to improve processing performance.
- Worked extensively with **AWS S3, Glue, Redshift, Athena, Kinesis, Lambda, EventBridge, SQS, and CloudWatch**.
- Developed and maintained ETL/ELT workflows supporting reliable downstream data consumption.
- Collaborated with stakeholders to translate requirements into production-ready **Python and data processing solutions**.
- Automated deployments using **CI/CD pipelines and infrastructure configuration**.

Python / Data Science Freelancer

UM6P-CS – Ben Guerir, Morocco | Mar 2021 – Aug 2021

- Collaborated with PhD research teams to translate mathematical and AI requirements into application solutions.
- Designed 2D/3D object detection approaches and developed feature extraction techniques from **LiDAR 3D point-cloud data**.
- Built feature-map sharing solutions to support model experimentation and performance analysis.
- Applied Python-based data processing and analytical techniques to research datasets.

Principal Developer

KASBAH PRODUCTIONS – Marrakech, Morocco | Jun 2018 – May 2019

- Developed and maintained web applications for commercial clients.
- Implemented application enhancements and backend functionality to improve reliability and delivery performance.
- Worked on application development, troubleshooting, and production support.

EDUCATION

Master – Intelligent and Decision-Making Systems | 2021
Faculty of Sciences, Fez, Morocco
Credential evaluated by World Education Services (WES)
Bachelor – Distributed Computer Systems | 2018
FST Marrakech, Morocco

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

- Building Modern Python Applications on AWS – 2021
- Neural Networks and Deep Learning – 2020